

# Kumbara by Sembol

• TESTNET

A self-custodial dollar piggy bank for Türkiye, opened with Face ID.

Live (Stellar TESTNET)

**kumbara.sembol.xyz**

Repository

**github.com/keyboard01/kumbara**

Deck date

**6 September 2026**

# Lira savings lose value; the dollar route has a wall in front of it.

## Inflation makes lira saving a losing move

Consumer prices in Türkiye rose **31.51% year on year** in August 2026.<sup>1</sup> Dollar-denominated saving is how ordinary households defend themselves.

## The crypto route exists but sheds users at every step

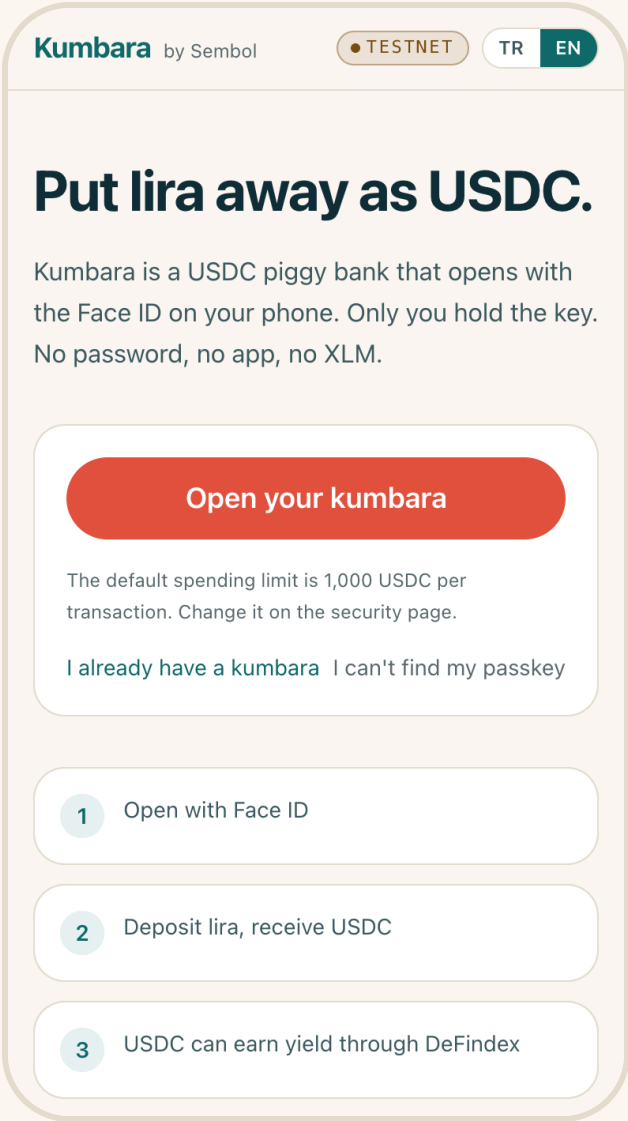
USDC on Stellar settles in seconds for fractions of a cent. Getting there means an extension or app install, twelve words to keep safe, XLM for gas bought somewhere else, and a phone you must never lose.

## The people who need hard-currency savings most are served least

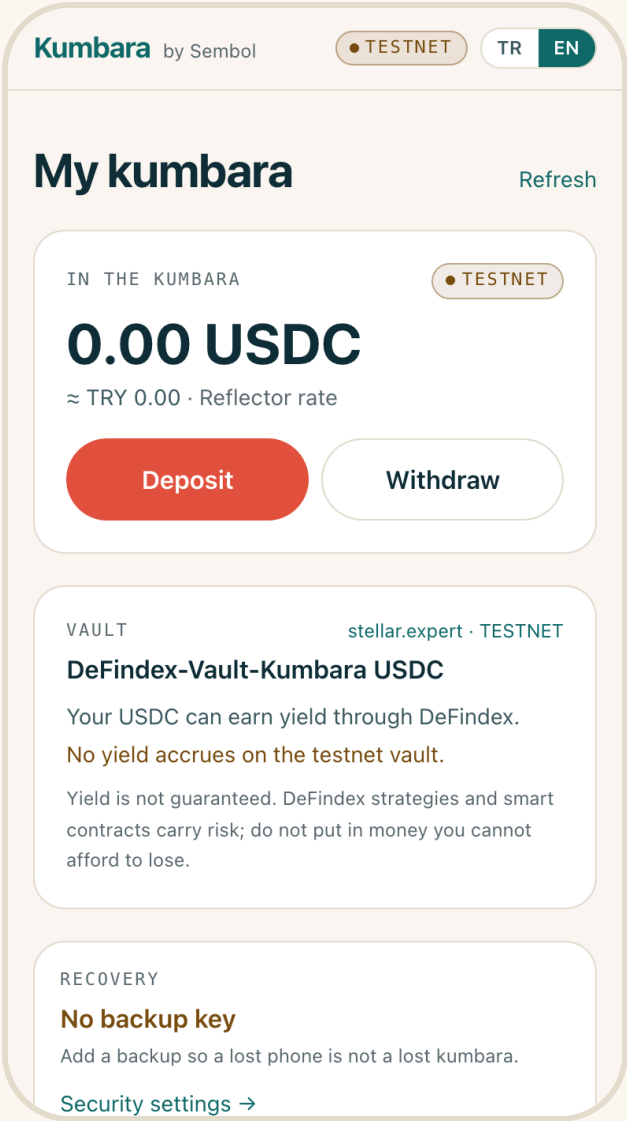
A bank-transfer habit and a phone with Face ID are what people have. That is the whole interface Kumbara asks for: no seed phrase, no XLM, no install.

<sup>1</sup> TÜİK (Turkish Statistical Institute), *Tüketici Fiyat Endeksi, Ağustos 2026*, news bulletin of 3 September 2026: CPI +31.51% versus August 2025 (monthly +1.84%). Bulletin index: veriportali.tuik.gov.tr; figure as published and as reported by Hürriyet Bigpara, [bigpara.hurriyet.com.tr/haberler/emtia-haberleri/agustos-ayi-enflasyonu-belli-oldu\\_ID102241364](https://bigpara.hurriyet.com.tr/haberler/emtia-haberleri/agustos-ayi-enflasyonu-belli-oldu_ID102241364).

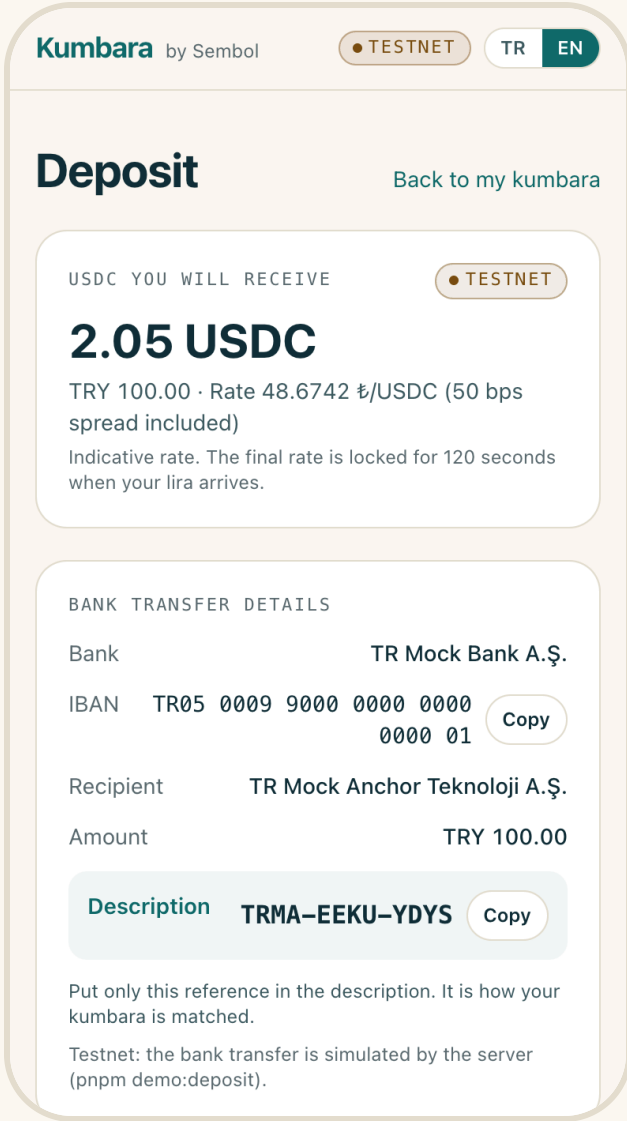
# Four screens. Tap to account: ~10 s. No seed phrase, no XLM, no install.



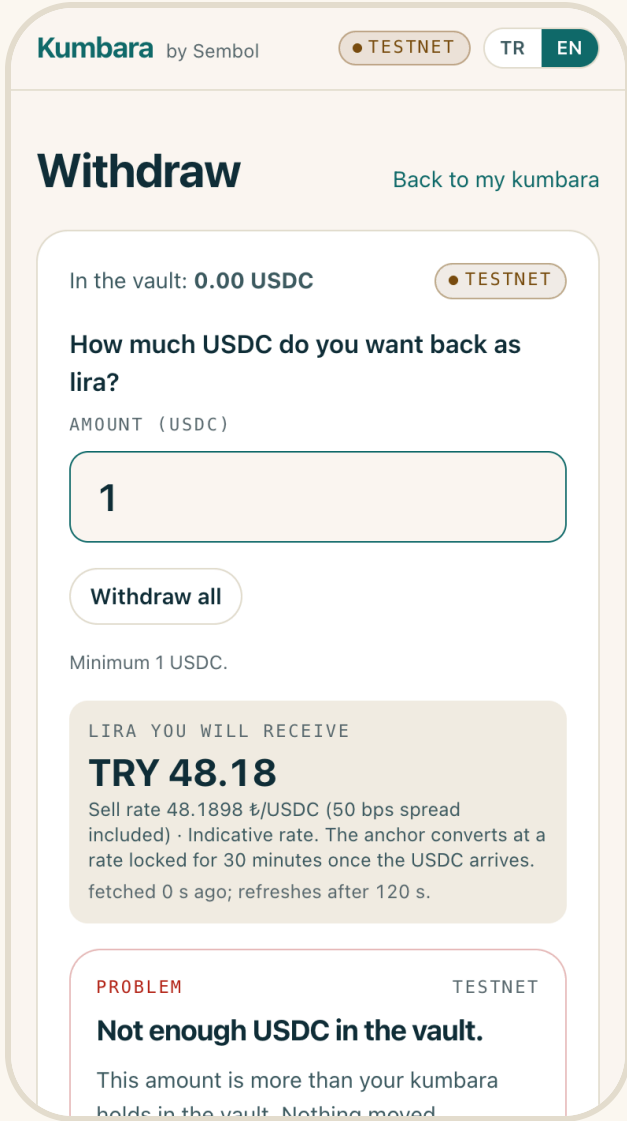
**Onboard.** One button; Face ID creates the key and the relay deploys an OpenZeppelin smart account. Measured 8–16 s tap to Savings in browser runs.



**Savings.** USDC in the DeFindex vault with its lira equivalent (Reflector rate), spending limit, recovery. Every figure carries the TESTNET badge.



**Deposit.** IBAN, recipient and a reference code, exactly like an online bank transfer; the anchor converts lira to USDC.



**Withdraw.** A sell quote in lira; two Face ID approvals; the anchor pays the visitor's own IBAN.

# Seven steps, one flow: anchor in front, DeFindex behind, a fee-sponsoring relay underneath.

- 1

The visitor opens an OpenZeppelin smart account with Face ID; the relay pays the fees.
- 2

A spending limit (1,000 USDC per transaction) is written on-chain in the background.
- 3

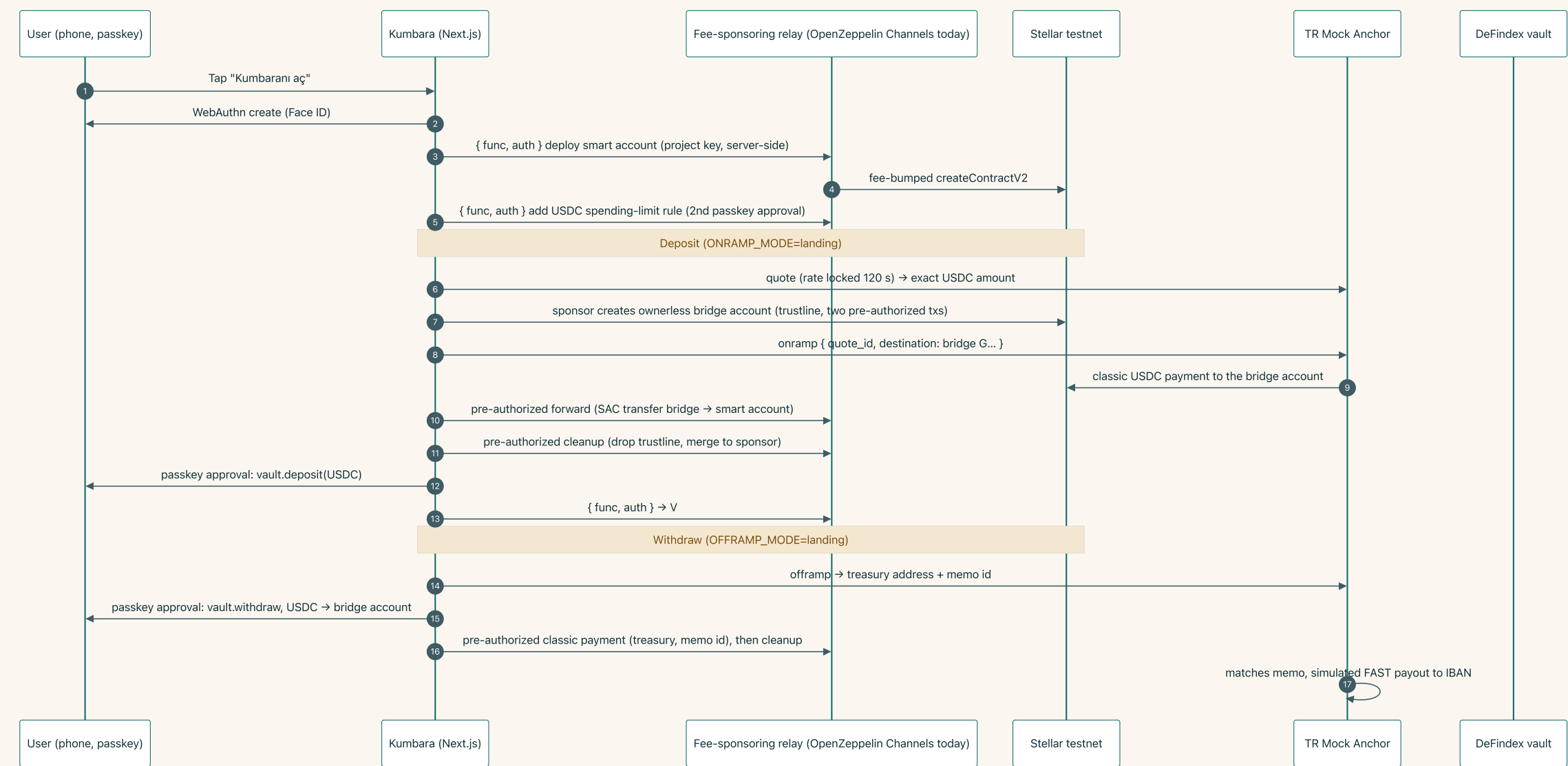
The anchor issues an IBAN and a reference; the visitor sends lira (simulated by the server on testnet).
- 4

The anchor converts the lira to USDC and pays an ownerless, pre-authorized bridge account.
- 5

The bridge account forwards the USDC to the kumbara and closes itself; nobody holds its key.
- 6

One Face ID puts the USDC into the DeFindex vault (no yield on testnet).
- 7

A withdrawal runs the path backwards: vault → bridge account → anchor → lira to the IBAN.



# Anchors pay classic accounts. Smart accounts are contracts. Ramp payouts could not reach them.

## The finding

The anchor's on-ramp rejects a contract (C...) destination and its watcher matches classic payments only: a payout cannot land in a smart account, and a smart account's Soroban transfer is never matched on the way back.

## The solution: ownerless, pre-authorized bridge accounts

A throwaway classic account whose only valid transactions are signed before it exists: the **forward** at sequence +1 (USDC to the user's smart account, the exact quoted amount) and the **cleanup** at sequence +2 (trustline dropped, reserves back to the sponsor). The master key is wiped in-process: **no key can sign anything else**. In reverse, the same primitive carries withdrawals to the anchor's treasury with the memo id.

10 / 10

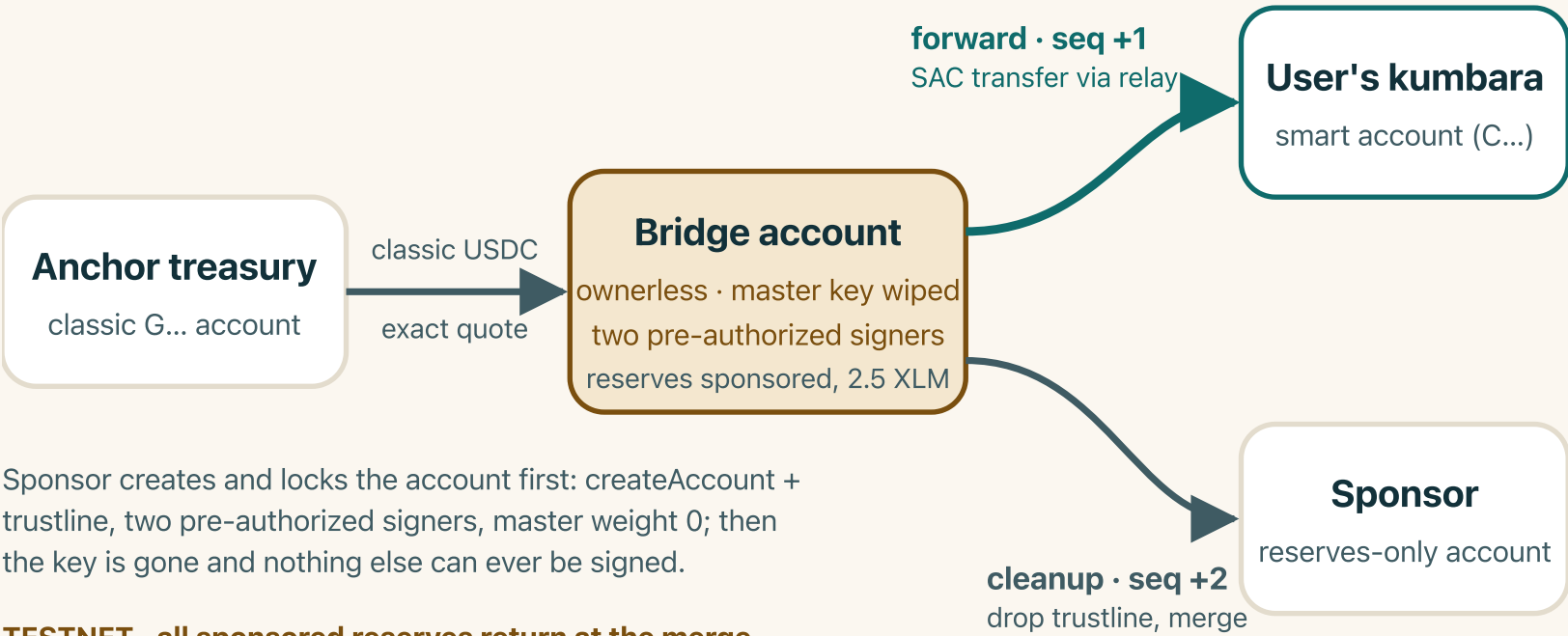
exact amount matches in ten scripted deposits

900

stroops net per deposit for the sponsor; reserves return at cleanup

4

signatures the bridge key ever produces, asserted by a unit test



Sponsor creates and locks the account first: createAccount + trustline, two pre-authorized signers, master weight 0; then the key is gone and nothing else can ever be signed.

TESTNET · all sponsored reserves return at the merge

| EVIDENCE (STELLAR TESTNET, STELLAR.EXPERT/EXPLORER/TESTNET/TX/...) | TRANSACTION         |
|--------------------------------------------------------------------|---------------------|
| Anchor pays the bridge account (classic payment)                   | 778215b3...e3c3a8f1 |
| Pre-authorized forward: USDC lands in the smart account            | f580e030...f22c9eab |
| Pre-authorized cleanup: trustline removed, account merged          | 6223c61c...7416291d |

Reported to the anchor team with the two fixes that would retire the bridge: SAC delivery to contract wallets and Soroban-aware matching on the way back. Full hashes in docs/anchor-notes.md.

# Listed building blocks, wired end to end. Kumbara writes no wallet, vault or strategy code.

| BUILDING BLOCK                                      | WHAT IT DOES IN THIS APP                                                                                                                                                                                    | ADDRESS (TESTNET, STELLAR.EXPERT)                   |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| OpenZeppelin smart accounts + spending-limit policy | Every kumbara is an audited smart account whose only signer is the visitor's passkey; a USDC-scoped rule caps 1,000 USDC per transaction on-chain. Via @sembol/passkey-react and stellar/smart-account-kit. | WASM 1b5f4534...97785a · policy CABXBYJN...2TIP5G   |
| DeFindex vault + hodl strategy                      | Deposits go through the vault's real invest path into its strategy and withdrawals unwind from it. The strategy is DeFindex's own hodl crate, built unmodified; no yield on testnet, none promised.         | vault CAT76PQM...MKSKL · strategy CDFLOE4H...JLDMQN |
| TR Mock Anchor · SEP-1/6/38 + Partner API           | The regulated lira side: IBAN and reference, quotes with a 120 s lock, on-ramps to the bridge account, off-ramps by memo, simulated FAST payouts. Issuer and endpoints read from its stellar.toml.          | USDC SAC CBIELTK6...XQDAMA                          |
| Fee-sponsoring relay (OpenZeppelin Channels today)  | Signed authorization entries go to our server route, which adds the project key and forwards to the relay; the visitor never sees XLM. Per-IP and per-booth caps, sponsor threshold.                        | channels.openzeppelin.com/testnet                   |
| Reflector                                           | The USD/TRY foreign-exchange oracle behind the lira equivalent on the Savings and stats pages (read by simulation, anchor rate as fallback).                                                                | CBKGPWGK...4DXMCJZC (mainnet oracle, read only)     |
| Soroswap <div>● FLAGGED</div>                       | Evaluated, not wired: the vault accepts the anchor's USDC directly. The router dry run is recorded for the day a swap is needed.                                                                            | router CCJUD55A...ZZE7BRD                           |

**Upstream:** the fix Sembol needed for passkey signing was merged into stellar/smart-account-kit and shipped in 0.6.2 

[PR link]

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# Everything on this slide is public, linkable and re-runnable.

Kumbara by Sembol

TESTNET

TR

EN

Kumbara live

live updated 1:17:47 AM

Public, read-only. Testnet: no real lira moves.

Since event start

All time

1

kumbaras opened

TESTNET

1

deposits completed

TESTNET

0

withdrawals completed

TESTNET

TRY 100.00

TRY in

TESTNET

TRY 0.00

TRY out

TESTNET

26.79

USDC

USDC in the vault (live)

TESTNET

**kumbara.sembol.xyz/stats**: headline numbers with a TESTNET badge, live feed, onboarding curve, timings. The *since event start* / *all time* toggle counts from 19 September 08:00 and shows all-time until then; seed and E2E accounts are excluded by default.

6

real-browser round trips in CI: onboard, deposit, withdraw, booth, stats, failure screens · every push and every 6 h against production

25

failure states, each a TR/EN screen with one action; every flow resumes from stored state

94 · 100 · 100 · 100

Lighthouse mobile on /stats: performance, accessibility, best practices, SEO

E2E · passing · latest production run 6 Sep 2026

GitHub Actions, workflow e2e.yml (repository badge available once the repo is public)

| ON-CHAIN EVIDENCE (STELLAR TESTNET)                                         | TRANSACTION         |
|-----------------------------------------------------------------------------|---------------------|
| Vault deployed through the DeFindex factory with the hodl strategy attached | dbf0d5b7...6faa8f96 |
| Deposit with invest=true moved USDC into the strategy (invest)              | b615e7dc...76b7f981 |
| Withdrawal unwound the shares from the strategy (divest)                    | 466f48d2...07a9f5ca |

Recorded fallback demo: docs/demo/round-trip.mp4, the full round trip on production, 2:31 with a caption per step.

PLACEHOLDER · FILL AFTER 20 SEPTEMBER

Hackathon onboarding: N accounts in 2 days (from /stats since event start, seed and E2E excluded).

# Non-custodial software in front of a licensed anchor.

- **Kumbara never holds lira, USDC or keys.** Every transfer is authorized by the visitor's own passkey on their own smart account; the bridge accounts have no owner.
- **The licensed anchor is the regulated party:** it holds the fiat, performs KYC and pays out only to the customer's own named IBAN.
- **No payments feature in Türkiye:** no send, no P2P, no merchant flow. Deposit into a vault and withdraw back to your own IBAN, nothing else.
- **Turkish AML rules (MASAK) shape the UX:** named-IBAN in and out, an on-chain per-transaction cap, anchor-side limits and no cash or third-party paths.
- **A Turkish regulatory opinion is budgeted in Tranche 1,** before any mainnet pilot; the mainnet path stays behind a capped, presenter-only flag until then.

# Small, shipping, already in the ecosystem.

**Ahmed** · lead

Instaward grantee. Author of [@sembol/passkey-react](#), the passkey wallet layer Kumbara runs on; contributor to [stellar/smart-account-kit](#). Built Kumbara end to end for this submission.

[github.com/keyboard01](#)

[LinkedIn URL]

**Ibo** · engineer

Engineering across the stack; ships the second tenant on the same infrastructure alongside Kumbara.

[LinkedIn URL] · [one-line bio to confirm]

**Atahan** · [role, if confirmed]

Deck, booth and visitor wrangling on event days, if confirmed.

[confirm participation] · [LinkedIn URL]

# Four tranches, each at most 90 days, each ending in a verifiable deliverable.

## TRANCHE 1 · ≤ 90 DAYS

### Ground the mainnet path

Turkish regulatory opinion; licensed-anchor production agreement signed; threat model and monitoring plan for the audit.

## TRANCHE 2 · ≤ 90 DAYS

### Mainnet audit pass

Audit of the smart-account, policy and bridge-account path; mainnet behind a capped, presenter-only flag; relay budgets and allowlists.

## TRANCHE 3 · ≤ 90 DAYS

### Kumbara public pilot

Public pilot with the licensed anchor on mainnet.

**On-chain metric:** DeFindex vault NAV from TRY deposits, target [X] USDC.

## TRANCHE 4 · ≤ 90 DAYS

### Native mobile SDK

Native mobile (Expo) SDK for the same passkey wallet layer; second-market groundwork.

## Expansion filter

The next market needs two things and nothing else: high inflation and an existing licensed Stellar anchor. Same app, same bridge, a different anchor in front.

## The ask

Integration Track award sized to the four tranches above: [amount], paid in XLM per the SCF schedule (10% acceptance, 20% MVP, 30% testnet, 40% mainnet on the committed metric).